

# Subhranil Deb

Université Paris Saclay, Orsay

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## EDUCATION

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**Institut Mathématiques de Jussieu - Paris Rive Gauche**

Oct 2026 - present

PhD in Mathematics, Double Kazhdan - Lusztig theory.

**Advisors:** Alexis Bouthier, Éric Vasserot

**Université Paris Saclay**

Sep 2025 - July 2026

M2 Analysis, Arithmetic, Geometry(AAG).

**Advisor:** Olivier Schiffmann

**Chennai Mathematical Institute**

Aug 2024 - May 2025

M1 Mathematics.

**Chennai Mathematical Institute**

Aug 2021 - June 2024

Bachelor of Science (Honours) in Mathematics and Computer Science.

## RESEARCH INTERESTS

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Geometric Langlands Program, Geometric Representation Theory, Algebraic Geometry

## PROJECTS/PRESENTATIONS

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- **M2 Thesis**

October 2025 - Present

**Instructor:** Olivier Schiffmann - Université Paris Saclay

– I am currently reading about cohomological integrality for symmetric quotient stacks and the cohomological Mackey formula following the papers by Lucien Hennecart.

- **Geometric Satake Equivalence**

May 2025 - July 2025

**Instructor:** Simon Riche - Université Clermont Auvergne

– I read the theory of perverse sheaves, equivariant derived categories leading up to the Geometric Satake Equivalence. The main reference I followed is the book “Perverse Sheaves and Applications to Representation Theory” by Achar.

- **Degree Bounds For Syzygies of Linearly Reductive Groups**

May 2024

**Instructor:** Manoj Kummini - Chennai Mathematical Institute

– As a part of the course **Commutative Algebra I**, I presented “Degree Bounds For Syzygies Of Invariants” by H. Dersken. Here is the [report](#).

- **Higher Dimensional Class Field Theory**

May 2024

**Instructor:** Sukhendu Mehrotra - Chennai Mathematical Institute

– As a part of the course **Geometric Class Field Theory**, I presented “Non-abelian class field theory for arithmetic surfaces” by Hoffman, Walter and Wiesend. Here is the [report](#).

- **Construction of Hilbert and Quot Schemes**

May 2024

**Instructor:** Mohit Upmanyu and Bivas Khan - Chennai Mathematical Institute

– As a part of the course **Algebraic Geometry II**, I presented the construction of Hilbert and Quot Schemes using Castelnuovo-Mumford regularity. Here is the [report](#).

- **Iwasawa Theory of Taelman Class Modules**

Dec 2023

**Instructor:** Anwesh Ray - Chennai Mathematical Institute

– I read the book Drinfeld Modules by Mihran Papikian for my course on Function Field Arithmetic. As a part of the course **Function Field Arithmetic**, I presented Higgins thesis: “Iwasawa theory of Taelman class modules”. Here is the [report](#).

- **Serre Duality on Compact Riemann Surfaces and Algebraic Curves**

May 2023 - June 2023

**Instructor:** Yogish Holla - TIFR, Mumbai

– I read R.R. Simha’s paper “The Riemann–Roch Theorem for Compact Riemann Surfaces” for a sheaf theoretic proof of Serre Duality.

– As a part of my final presentation, I read the first 3 chapters of Diamond-Shurman’s text on modular forms and presented the dimension formula as an application of the Riemann–Roch Formula.

## TALKS

Perverse Sheaves and The Geometric Satake Equivalence, CMI Student Seminars([website](#)), August 2025  
BGG Resolution and Higher Ext Groups in Category  $\mathcal{O}$ , Lie Algebra Representations Course, June 2025  
Quasicoherent Sheaves on Toric Varieties, Toric Varieties Course, April 2025  
Grothendieck local duality, Commutative Algebra II Course, April 2025  
Clifford Algebras, Pin and Spin Groups, Lie Groups and Lie Algebra Course, Nov 2024  
Galois Representations and Iwasawa Theory, CMI Student Seminars([website](#)), Jan 2024  
Modular forms and Riemann-Roch Theorem, VSRP presentation, July 2023

## ACHIEVEMENTS

2025      Awarded the Sophie Germain Scholarship for admission in the M2 AAG program by FMJH, Université Paris Saclay  
2024      Selected for a fully funded research internship by [ReLaX](#) under Prof. Simon RICHE, Université Clermont Auvergne  
2024      Awarded Scholarship for admission in the MSc Mathematics program at CMI, Chennai Mathematical Institute  
2023      Selected for TIFR VSRP (Visiting Students' Research Program), Tata Institute of Fundamental Research (TIFR), Mumbai  
2022      Selected for Madhava Nurture camp, S.P. College & HBCSE  
2019      Qualified Regional Mathematics Olympiad(RMO), HBCSE  
2019      Qualified MTRP, Indian Statistical Institute, Kolkata

## CONFERENCES/WORKSHOPS

- Derived Algebraic Geometry, CoHAs and Operads      June 2026  
Université Angers
- Enumerative geometry, rationality, derived categories and interactions      June 2026  
Université Paris Saclay
- Mathematics Inspired by Physics      June 2026  
IHES
- CARE - Collaborations in Algebra, Representation Theory and Ethics      October 2025  
ENS Lyon
- Geometry and Representation Theory around the Langlands Programs      July 2025  
Université Clermont Auvergne
- Introduction to the p-adic local Langlands program      September 2024  
Indian Institute of Technology Jammu
- Derived categories and semi-orthogonal decompositions      June 2024  
Chennai Mathematical Institute
- Rational Points on Modular Curves      September 2023  
International Centre of Theoretical Sciences(Online)

## TEACHING EXPERIENCE

I have been a teaching assistant for the following courses at CMI:

Topology (Instructor: Prof. Aditya Karnataki)  
Algebra I (Instructor: Prof. Aditya Karnataki)  
Topology (Instructor: Prof. Parameswaran Sankaran)  
Analysis 3 (Instructor: Prof. Srinivasan Vasanth)  
Analysis 2 (Instructor: Prof. Upendra Kulkarni)  
Advanced Programming in Python (Instructor: Prof. Samir Datta)  
Algebra 1 (Instructor: Prof. Upendra Kulkarni)  
Introduction To Programming In Haskell (Instructor: Prof. S P Suresh)